ARC-AGI Milestone A

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1 INTRODUCTION

For Milestone A, my primary goals were to understand how ARC-AGI problems are represented and walk-through and complete as many examples as possible. In this journal, I will describe how I reasoned through examples 0520fde7, 1cf80156, and 28e73c20, how my agent performed, and what challenges I met as I worked through the problems.

2 HOW WOULD YOU, AS A HUMAN, REASON THROUGH ARC-AGI PROBLEMS?

Across all three problems, I:

- identify the structure,
- look for regularities that describe the behavior, and
- decide on the transformation that explain the scenario.

2.1 My Approach to 0520fde7

In this problem, each training input is a 3x7 grid with a vertical 3x1 column of 5s in the middle and a pattern of 1s and 0s on either side. The outputs are 3x3 grids containing 2s in the positions of overlap in the input and on either side of the 5s.

I treated the central column of 5s as the separator that doesn't change, which allowed me to look at 3 3x3 grids. I split up the input into 3 sections, a left, right, and middle. I then analyzed the output to see how it might be the result of the input. When looking at the training data I could see that you put a 2 in the output where ever the 2 3x3 grids in the input had overlap.

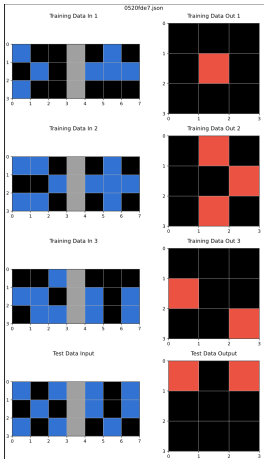


Figure 1—0520fde7 - Overlap

2.2 My Approach to 1cf80156

In this problem, the inputs are large grids of mostly zeros with a single cluster of non-zero color and the outputs are smaller grids that container only the non-zero clusters. For example, one input has a 10x12 input grid with a cluster of 2s, and an 4x4 output grid of only those 2s.

It easy to identify pretty quickly that this transformation is essentially cropping out the cluster. Once the non-zero region is identified, you can draw the smallest rectangle that contains each non-zero number. Then, you can pop out that section.

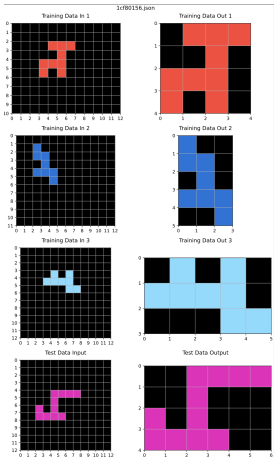


Figure 2—1cf80156 - Cropping

2.3 My Approach to 28e73c20

In this problem, every input is a grid of 0s and the output is a same size grid of a recursive spiral and filled with 3s and 0s. The spiral gets bigger as the size of the input grid gets bigger.

This was a tough problem because the input shares very little information beyond grid size. The output is generated purely from the size of the grid with the same pattern. There's a border of 3s with paths of 0s, and the spiral moves inward toward the center. The pattern is a base spiral, then with each larger input grid, we get a new layer of 3s with a path of 0s.

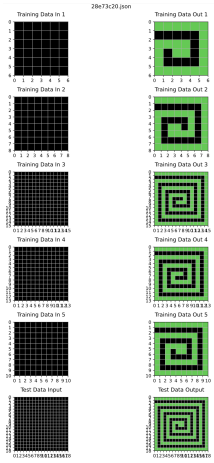


Figure 3—28e73c20 - Spiral

3 HOW DO YOU EXPECT YOU WILL DESIGN AN AGENT TO APPROACH THESE PROBLEMS?

My agent uses a strategy that combines a hand-written bank of transformation rules while also learning parameters from the training pairs. My approach was to recognize when one of these known transformations fits a problem, rather than trying to invent entirely new transformations.

In all three problems, the agent is able to select from a set of rules whose behavior can be understood, ie find overlaps, cropping, and create a spiral.

3.1 Agent Approach to 0520fde7

I expected my agent to use a specific rule for "finding the overlap" implement in the rule bank. This rule:

- checks that each training input is 3 rows high and has a central column of 5s,
- confirms that the outputs are always 3x3 and use the color 2 when the overlap is found, and
- treats the 3x7 as being mapped onto a 3x3.

Within this rule, the agent learned which subsets of input columns correspond to each output column by comparing where 1s appeaer in the training inputs to where 2s appear in the training outputs. Once those mappings are found across training pairs, the agent applies the same compression to the test input, producing the 3x3 output gris with 2s in the overlapping areas.

3.2 Agent Approach to 1cf80156

For this problem, the agent relies on generic cropping that is in the rule bank. This rule:

- finds all non-zero cells in the input grids,
- computes the minimal bounding box that contains them, and
- returns the subgrid inside that bounding box as the output.

Since the training pairs show a single cluster in each input and no recoloring or rotations, the training output is produced exactly. The agent validates this by applying the cropping rule to each training input and checking whether the result matches the provides output. If all match, the rule is considered passing.

3.3 Agent Approach to 28e73c20

For this problem, the agent uses a fully generative spiral rule. This rule:

- requires that the input grid be square and contain only 0s,
- chooses a small base pattern of 3s and 0s for the smallest sizes, and
- repeatedly grows the pattern by wrapping it with a new frame of 3s and 0s path.

The agent tests this rule by generating an output of the same size as each training input and comparing it to the corresponding training output. If all generated patterns match, the rule is accepted as the correct interpretation of the problem.

4 WHAT DO YOU ANTICIPATE YOUR BIGGEST CHALLENGES IN DESIGNING YOUR AGENT WILL BE AND WHY?

Working through these problems I realize several challenges I would need to overcome. From the highest, much of this milestone seemed like brute force which may or may not have generalized well.

First, for problems such as 0520fde7 my agent would need to learn how to find the overlaps and that it is in fact overlap, that the middle column of 5s is a kind of pivot point, and this must generalize to a test input. Designing these builders was more difficult than I expected.

Second, for problems like 1cf80156, my agent need to learn and understand what the full non-zero pattern was such that it didn't miss any portions. I could see this becoming more challenging with multiple clusters or if there are other transformations such as rotations before cropping.

Third, for recursive problems like 28e73c20, the difficulty is learning the spiral from essentially zero information and the ability to create spirals in a dynamic way, i.e. as the grid grows so does the spiral. Small mistakes in the base cases and cause other large grids to fail rapidly.

Project-wide, I expect continued challenges with brute-force methods as the problems get harder and the ability to generalize becomes more necessary. If rules are too specific I run the risk of missing test cases. As I progress through the milestones, I will need to ensure my agent takes a more general approach.